Solution of one-dimensional wave equation by finite difference approach.

Consider the one dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \rightarrow (1)$$

Suppose we wish to solve equation (1) for $0<x<a$ and $t>0$ under the

Boundary conditions:

$$u(0,t) = \phi(t) \; ; \; u(a,t) = \psi(t) \; , \; t \geq 0$$

where $\phi(t)$ and $\psi(t)$ are specified functions of t $(t \geq 0)$ and the initial conditions:

Initial position $u(x,0) = f(x) \; ; \; 0<x<a$

Initial velocity of the string $\frac{\partial u}{\partial t}(x,0) = g(x) \; , \; 0<x<a$

where $f(x)$ and $g(x)$ are specified functions of x (for $0<x<a$)

(1) $\Rightarrow$

$$\frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2} = c^2 \left[\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} \right]$$

$$u_{i,j+1} - 2u_{i,j} + u_{i,j-1} = \frac{c^2 k^2}{h^2} \left[u_{i+1,j} - 2u_{i,j} + u_{i-1,j} \right]$$

Taking $\beta^2 = \frac{c^2 k^2}{h^2}$, we get

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$$u_{i,j+1} = \beta^2(u_{i+1,j} + u_{i-1,j}) - 2\beta^2 u_{i,j} + 2u_{i,j} - u_{i,j-1}$$

$$u_{i,j+1} = \beta^2(u_{i+1,j} + u_{i-1,j}) + 2(1-\beta^2)u_{i,j} - u_{i,j-1}$$

If $\beta = 1$, then

$\boxed{\beta = 1 \Rightarrow k = \frac{h}{c}}$

$$\boxed{u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}} \rightarrow (2)$$

The above equation is called 3-level explicit scheme for 1-D wave equation

NOTE:- the method is stable when $\beta \leq 1$ and unstable when $\beta > 1$.

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(1) By using the explicit three level formula, Solve the wave equation $\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}$ subject to the boundary conditions

$u(0,t) = u(4,t) = 0$, $t \geq 0$ and the initial conditions

$u(x,0) = x(4-x)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $0 < x < 4$. Take

$h = 1$ and $k = \frac{1}{2}$ and compute $u(x,t)$ for $0 < t \leq 2$.

Soln:- Given $h = 1$, $k = \frac{1}{2}$, $c^2 = 4$, $\beta = \frac{ck}{h} = \frac{2(1/2)}{1} = 1$

Given initial condition

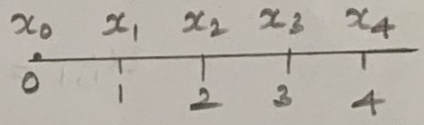
$u(x,0) = x(4-x)$

$u(x_i, 0) = x_i(4-x_i)$

$u_{i,0} = ih(4-ih)$

$u_{i,0} = i(4-i)$ for $i = 1, 2, 3$

$u_{1,0} = 3$, $u_{2,0} = 4$, $u_{3,0} = 3$



Boundary conditions

$u(0,t) = u(x_0, t_j) = u_{0,j} = 0$ for $j = 0, 1, 2, 3$

$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = 0$

$u(4,t) = u(x_4, t_j) = u_{4,j} = 0$ for $j = 0, 1, 2, 3$

$\Rightarrow u_{4,0} = u_{4,1} = u_{4,2} = u_{4,3} = u_{4,4} = 0$

By the three level explicit formula, we have

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \rightarrow (1)$$

(2)

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For the first time level, we put $j=0$ in (1)

$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1} \quad \rightarrow (2)$$

We note that the initial velocity given is $\frac{\partial u}{\partial t} = 0$

Also by central difference $\quad \frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0$

$$\Rightarrow u_{i,j+1} - u_{i,j-1} = 0$$

$$\Rightarrow u_{i,j+1} = u_{i,j-1}$$

$$\Rightarrow u_{i,1} = u_{i,-1} \quad \text{(Taking } j=0\text{)}$$

$\therefore$ (2) becomes

$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,1}$$

$$\Rightarrow 2u_{i,1} = u_{i+1,0} + u_{i-1,0}$$

$$u_{i,1} = \frac{u_{i+1,0} + u_{i-1,0}}{2}$$

For $i=1, 2, 3$, we get

$$u_{1,1} = \frac{u_{2,0} + u_{0,0}}{2} = \frac{1}{2}[4+0] = 2$$

$$u_{2,1} = \frac{u_{3,0} + u_{1,0}}{2} = \frac{1}{2}[3+3] = 3$$

$$u_{3,1} = \frac{u_{4,0} + u_{2,0}}{2} = \frac{1}{2}[0+4] = 2$$

The above expressions constitute the solution at first time level for $t = k = \frac{1}{2}$

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For the second time level, we put $j=1$ in (1)

$\therefore u_{i,2} = u_{i+1,1} + u_{i-1,1} - u_{i,0}$

for $i=1,2,3$, we get

$$u_{1,2} = u_{2,1} + u_{0,1} - u_{1,0} = 3+0-3 = 0$$

$$u_{2,2} = u_{3,1} + u_{1,1} - u_{2,0} = 2+2-4 = 0$$

$$u_{3,2} = u_{4,1} + u_{2,1} - u_{3,0} = 0+3-3 = 0$$

The above expressions constitute the solution at second time level for $t = 2k = 2(\frac{1}{2}) = 1$

For the third time level, we put $j=2$ in (1)

$\therefore u_{i,3} = u_{i+1,2} + u_{i-1,2} - u_{i,1}$

For $i=1,2,3$, we get

$$u_{1,3} = u_{2,2} + u_{0,2} - u_{1,1} = 0+0-2 = -2$$

$$u_{2,3} = u_{3,2} + u_{1,2} - u_{2,1} = 0+0-3 = -3$$

$$u_{3,3} = u_{4,2} + u_{2,2} - u_{3,1} = 0+0-2 = -2$$

The above expression constitute the solution at third time level for $t = 3k = 3(\frac{1}{2}) = \frac{3}{2}$.

For the Fourth time level, Put $j=3$ in (1)

$\therefore u_{i,4} = u_{i+1,3} + u_{i-1,3} - u_{i,2}$

For $i=1,2,3$ we get

$$u_{1,4} = u_{2,3} + u_{0,3} - u_{1,2} = -3+0-0 = -3$$

$$u_{2,4} = u_{3,3} + u_{1,3} - u_{2,2} = -2-2-0 = -4$$

$$u_{3,4} = u_{4,3} + u_{2,3} - u_{3,2} = 0-3-0 = -3.$$

The above values given the solution at 4th time level $t=2$

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② Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions

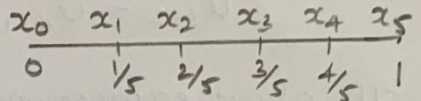
$u(0,t) = 0 = u(1,t)$; $t \geq 0$ and initial conditions

$u(x,0) = \sin \pi x$, $\frac{\partial u}{\partial t}(x,0) = 0$, $0 < x < 1$ taking $h = k = 0.2$

at $t = 0$.

Soln: Given $c^2 = 1$, $h = k = 0.2$ and $k = \frac{h}{c} \Rightarrow \beta = 1$

By the initial condition



$u(x,0) = \sin \pi x$

$\Rightarrow u(x_i, 0) = \sin \pi x_i = \sin \pi (ih) = \sin\left(\frac{i\pi}{5}\right)$ $\left(\because h = \frac{1}{5}\right)$

$u_{i,0} = \sin\left(\frac{i\pi}{5}\right)$ for $i = 1, 2, 3, 4$

$u_{1,0} = \sin\left(\frac{\pi}{5}\right) = 0.5877$

$u_{2,0} = \sin\left(\frac{2\pi}{5}\right) = 0.9510$

$u_{3,0} = \sin\left(\frac{3\pi}{5}\right) = 0.9510$

$u_{4,0} = \sin\left(\frac{4\pi}{5}\right) = 0.5877$

Boundary condition:

$u(0,t) = u(x_0, t_j) = u_{0,j} = 0$

$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = u_{0,5} = \cdots = 0$

$u(1,t) = u(x_5, t_j) = u_{5,j} = 0$

$\Rightarrow u_{5,0} = u_{5,1} = u_{5,2} = \cdots u_{5,5} = \cdots = 0$

By the three level explicit formula, we have

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1} \rightarrow (1)$$

For the first time level, we take $j = 0$ in (1).

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$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1} \rightarrow (2)$$

We note that the initial velocity given is $\frac{\partial u}{\partial t} = 0$

$$\therefore \quad \frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0$$

$$\Rightarrow u_{i,j+1} - u_{i,j-1} = 0$$

$$\Rightarrow u_{i,j+1} = u_{i,j-1}$$

$$\Rightarrow u_{i,1} = u_{i,-1} \quad \text{for } (j=0)$$

$\therefore$ (2) becomes

$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,1}$$

$$2u_{i,1} = u_{i+1,0} + u_{i-1,0}$$

$$u_{i,1} = \frac{u_{i+1,0} + u_{i-1,0}}{2}$$

For $i = 1, 2, 3, 4$, we get

$$u_{1,1} = \frac{1}{2}[u_{0,0} + u_{2,0}] = \frac{1}{2}[0 + 0.9510]$$

$$u_{1,1} = 0.4755$$

$$u_{2,1} = \frac{1}{2}[u_{1,0} + u_{3,0}] = \frac{1}{2}[0.5877 + 0.9510]$$

$$= 0.7693$$

$$u_{3,1} = \frac{1}{2}[u_{2,0} + u_{4,0}] = \frac{1}{2}[0.9510 + 0.5877]$$

$$= 0.7693$$

$$u_{4,1} = \frac{1}{2}[u_{3,0} + u_{5,0}] = \frac{1}{2}[0.9510 + 0]$$

$$= 0.4755$$

1st time level $t = 1/5$ or $t = 0.2$

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For the second time level, we put $j=1$ in (1),

$$u_{i,2} = u_{i-1,1} + u_{i+1,1} - u_{i,0}$$

for $i = 1, 2, 3, 4$

$$u_{1,2} = u_{0,1} + u_{2,1} - u_{1,0} = 0 + 0.7693 - 0.5877 = 0.1816$$

$$u_{2,2} = u_{1,1} + u_{3,1} - u_{2,0} = 0.4755 + 0.7693 - 0.9510 = 0.2938$$

$$u_{3,2} = u_{2,1} + u_{4,1} - u_{3,0} = 0.7693 + 0.4755 - 0.9510 = 0.2938$$

$$u_{4,2} = u_{3,1} + u_{5,1} - u_{4,0} = 0.7693 + 0 - 0.5877 = 0.1816$$

The above values constitute the second time level at $t = 0.4$.

For the third time level, we put $j=2$ in (1)

$$u_{i,3} = u_{i-1,2} + u_{i+1,2} - u_{i,1}$$

for $i = 1, 2, 3, 4$

$$u_{1,3} = u_{0,2} + u_{2,2} - u_{1,1} = 0 + 0.2938 - 0.4755 = -0.1816$$

$$u_{2,3} = u_{1,2} + u_{3,2} - u_{2,1} = 0.1816 + 0.2938 - 0.7693 = -0.2939$$

$$u_{3,3} = u_{2,2} + u_{4,2} - u_{3,1} = 0.2938 + 0.1816 - 0.7693 = -0.2939$$

$$u_{4,3} = u_{3,2} + u_{5,2} - u_{4,1} = 0.2938 + 0 - 0.4755 = -0.1816$$

The above values constitute the third time level at $t = 0.6$.

For the Fourth time level, we put $j = 3$ in (1)

$$u_{i,4} = u_{i-1,3} + u_{i+1,3} - u_{i,2}$$

for $i = 1, 2, 3, 4$

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$$u_{1,4} = u_{0,3} + u_{2,3} - u_{1,2} = 0 + (-0.2939) - 0.1816 = -0.4755$$

$$u_{2,4} = u_{1,3} + u_{3,3} - u_{2,2} = -0.1816 - 0.2939 - 0.2938 = -0.7693$$

$$u_{3,4} = u_{2,3} + u_{4,3} - u_{3,2} = -0.2939 - 0.1816 - 0.2938 = -0.7693$$

$$u_{4,4} = u_{3,3} + u_{5,3} - u_{4,2} = -0.2939 + 0 - 0.1816 = -0.4755$$

The above values constitute the fourth time level at $t=0.8$

For the fifth time level, we put $j=4$ in (1)

$$u_{i,5} = u_{i-1,4} + u_{i+1,4} - u_{i,3}$$

for $i = 1, 2, 3, 4$

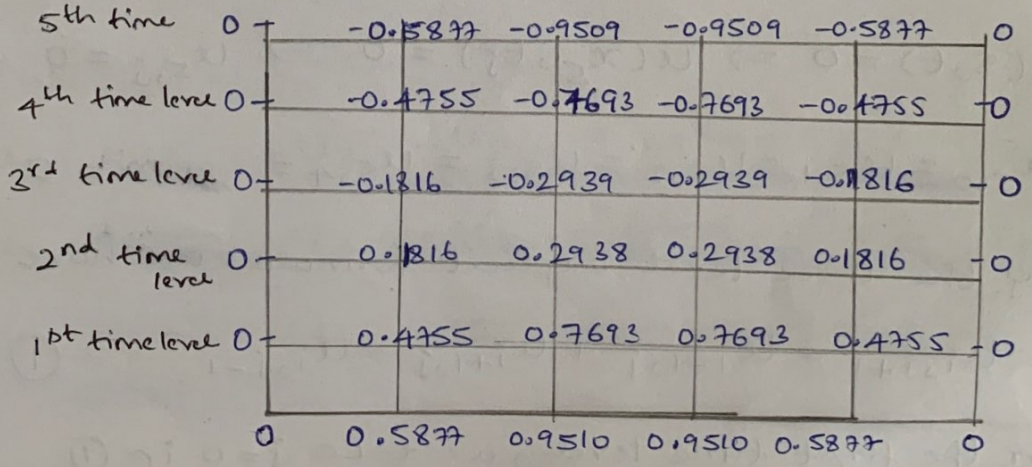
$$u_{1,5} = u_{0,4} + u_{2,4} - u_{1,3} = 0 + (-0.7693) - (-0.1816) = -0.5877$$

$$u_{2,5} = u_{1,4} + u_{3,4} - u_{2,3} = -0.4755 - 0.7693 - (-0.2939) = -0.9509$$

$$u_{3,5} = u_{2,4} + u_{4,4} - u_{3,3} = -0.7693 - 0.4755 - (-0.2939) = -0.9509$$

$$u_{4,5} = u_{3,4} + u_{5,4} - u_{4,3} = -0.7693 + 0 - (-0.1816) = -0.5877$$

The above values constitute the fifth time level at $t=1$



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3) Evaluate the pivotal values of the equation

$u_{tt} = 16 u_{xx}$ taking $h=1$ upto $t=1$.

The boundary conditions are $u(0,t) = u(5,t) = 0$

initial conditions $u(x,0) = x^2(5-x)$ and

$$\frac{\partial u}{\partial t}(x,0) = 0.$$

Soln: given $c^2 = 16$, $h = 1$, $\beta = \frac{ck}{h^2} \Rightarrow 1 = \frac{4(k)}{1}$

$$\Rightarrow k = \frac{1}{4}$$

By the initial condition, we have

$$u(x,0) = x^2(5-x)$$

$$\Rightarrow u(x_i, 0) = (x_i)^2(5-x_i)$$

$$u_{i,0} = (ih)^2(5-ih) \quad (\because h=1)$$

$$u_{i,0} = i^2(5-i) \quad \text{for } i = 1,2,3,4$$

$\therefore u_{1,0} = 4$, $u_{2,0} = 12$, $u_{3,0} = 18$, $u_{4,0} = 16$

Boundary condition:

x_0	x_1	x_2	x_3	x_4	x_5
0	1	2	3	4	5

$$u(0,t) = u(x_0, t_j) = u_{0,j} = 0$$

$$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = u_{0,5} = \cdots = 0$$

$$u(5,t) = 0 \Rightarrow u(x_5, t_j) = 0 \Rightarrow u_{5,j} = 0 \quad \text{for } j = 1,2,3,$$

$$u_{5,0} = u_{5,1} = u_{5,2} = u_{5,3} = u_{5,4} = \cdots = 0$$

By the three level explicit formulas, we have

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1} \rightarrow (1)$$

For the first time level, we take $j=0$ in (1)

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$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1} \rightarrow (2)$$

We note that the initial velocity given is $\frac{\partial u}{\partial t} = 0$

$$\therefore \quad \frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0$$

$$\Rightarrow u_{i,1} = u_{i,-1} \quad (\text{for } j=0)$$

$\therefore$ (2) becomes

$$u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,1}$$

$$\Rightarrow u_{i,1} = \frac{u_{i+1,0} + u_{i-1,0}}{2}$$

For $i = 1, 2, 3, 4$ we get

$$u_{1,1} = \frac{1}{2}[u_{0,0} + u_{2,0}] = \frac{1}{2}[0+12] = 6$$

$$u_{2,1} = \frac{1}{2}[u_{1,0} + u_{3,0}] = \frac{1}{2}[4+18] = 11$$

$$u_{3,1} = \frac{1}{2}[u_{2,0} + u_{4,0}] = \frac{1}{2}[12+16] = 14$$

$$u_{4,1} = \frac{1}{2}[u_{3,0} + u_{5,0}] = \frac{1}{2}[18+0] = 9$$

} First time level at $t = 1/5$

For the second time level, we put $j = 1$ in (1)

$$\Rightarrow u_{i,2} = u_{i-1,1} + u_{i+1,1} - u_{i,0}$$

For $i = 1, 2, 3, 4$ we get

$$u_{1,2} = u_{0,1} + u_{2,1} - u_{1,0} = 0 + 11 - 4 = 7$$

$$u_{2,2} = u_{1,1} + u_{3,1} - u_{2,0} = 6 + 14 - 12 = 8$$

$$u_{3,2} = u_{2,1} + u_{4,1} - u_{3,0} = 11 + 9 - 18 = 2$$

$$u_{4,2} = u_{3,1} + u_{5,1} - u_{4,0} = 14 + 0 - 16 = -2$$

} Second time level $t = 2/5$

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For the third time level, we put $j=2$ in (1)

$$u_{i,3} = u_{i-1,2} + u_{i+1,2} - u_{i,1}$$

For $i = 1, 2, 3, 4$

$$u_{1,3} = u_{0,2} + u_{2,2} - u_{1,1} = 0 + 8 - 6 = 2$$
$$u_{2,3} = u_{1,2} + u_{3,2} - u_{2,1} = 7 + 2 - 11 = -2$$
$$u_{3,3} = u_{2,2} + u_{4,2} - u_{3,1} = 8 + (-2) - 14 = -8$$
$$u_{4,3} = u_{3,2} + u_{5,2} - u_{4,1} = 2 + 0 - 9 = -7$$

} 3rd time level at $t = 3/5$

For the Fourth time level, we put $j=3$ in (1)

$$u_{i,4} = u_{i-1,3} + u_{i+1,3} - u_{i,2}$$

For $i = 1, 2, 3, 4$

$$u_{1,4} = u_{0,3} + u_{2,3} - u_{1,2} = 0 + (-2) - 7 = -9$$
$$u_{2,4} = u_{1,3} + u_{3,3} - u_{2,2} = 2 - 8 - 8 = -14$$
$$u_{3,4} = u_{2,3} + u_{4,3} - u_{3,2} = -2 - 7 - 2 = -11$$
$$u_{4,4} = u_{3,3} + u_{5,3} - u_{4,2} = -8 + 0 - (-2) = -6$$

} 4th time level at $t = 4/5$

For the fifth time level, we put $j=4$ in (1)

$$u_{i,5} = u_{i-1,4} + u_{i+1,4} - u_{i,3}$$

For $i = 1, 2, 3, 4$

$$u_{1,5} = u_{0,4} + u_{2,4} - u_{1,3} = 0 + (-14) - 2 = -16$$
$$u_{2,5} = u_{1,4} + u_{3,4} - u_{2,3} = -9 - 11 - (-2) = -18$$
$$u_{3,5} = u_{2,4} + u_{4,4} - u_{3,3} = -14 - 6 - (-8) = -12$$
$$u_{4,5} = u_{3,4} + u_{5,4} - u_{4,3} = -11 + 0 - (-7) = -4$$

} 5th time level at $t = 1$

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4) By using explicit three-level formulas, solve the wave equation $4\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ subject to the conditions

$$u(0,t) = u(1,t) = 0, \quad t \geq 0$$

$$u(x,0) = \begin{cases} \frac{5x}{3}, & 0 \leq x \leq 3/5 \\ \frac{5(1-x)}{2}, & 3/5 < x < 1 \end{cases}$$

$\frac{\partial u}{\partial t}(x,0) = 0$. Taking $h = 1/5$ carry out computations upto $t = 2/5$.

Soln:- Given $h = \frac{1}{5}$, $c = 2$, $\quad k = \frac{h}{c} = \frac{\frac{1}{5}}{2} = \frac{1}{10}$ (i.e. $\beta = 1$)

using boundary conditions:

x_0	x_1	x_2	x_3	x_4	x_5
0	1/5	2/5	3/5	4/5	1

$$u(0,t) = u(x_0, t_j) = u_{0,j} = 0$$

$$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = \cdots = 0$$

$$u(1,t) = u(x_5, t_j) = u_{5,j} = 0$$

$$\Rightarrow u_{5,0} = u_{5,1} = u_{5,2} = u_{5,3} = u_{5,4} = \cdots = 0$$

Initial condition gives

$$u(x,0) = u(x_i, 0) = \begin{cases} \frac{5x_i}{3}, & 0 < x_i \leq 3/5 \\ \frac{5(1-x_i)}{2}, & 3/5 < x_i < 1 \end{cases}$$

$$u_{i,0} = \begin{cases} \frac{5ih}{3}, & i = 1, 2, 3 \\ \frac{5(1-ih)}{2}, & i = 4 \end{cases}$$

$$u_{i,0} = \begin{cases} \frac{i}{3}, & i = 1, 2, 3 \\ \frac{5(1-\frac{i}{5})}{2}, & i = 4 \end{cases}$$

(7)

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$\therefore u_{1,0} = \frac{1}{3}, \quad u_{2,0} = \frac{2}{3}, \quad u_{3,0} = 1, \quad u_{4,0} = \frac{5}{2}\left(1 - \frac{1}{5}\right)$

$$u_{4,0} = \frac{1}{2}$$

Also $\frac{\partial u}{\partial t} = 0$

$$\Rightarrow \frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0$$

$$\Rightarrow u_{i,j+1} = u_{i,j-1}$$

for $j = 0$, $u_{i,1} = u_{i,-1}$

By the three-level explicit formula

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \rightarrow (1)$$

For the first time level, we put $j = 0$ in (1)

$$\Rightarrow u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1}$$

$$\Rightarrow u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,1}$$

$$\Rightarrow u_{i,1} = \frac{u_{i+1,0} + u_{i-1,0}}{2}$$

for $i = 1, 2, 3, 4$

$$u_{1,1} = \frac{u_{2,0} + u_{0,0}}{2} = \frac{1}{2}\left(\frac{2}{3} + 0\right) = \frac{1}{3}$$

$$u_{2,1} = \frac{u_{3,0} + u_{1,0}}{2} = \frac{1}{2}\left(1 + \frac{1}{3}\right) = \frac{2}{3}$$

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$$u_{3,1} = \frac{u_{4,0} + u_{2,0}}{2} = \frac{1}{2}\left(\frac{1}{2} + \frac{2}{3}\right) = \frac{7}{12}$$

$$u_{4,1} = \frac{u_{5,0} + u_{3,0}}{2} = \frac{1}{2}(0+1) = \frac{1}{2}$$

The above values constitute the first time level solution at $t = k = \frac{1}{10}$.

For the second time level, we put $j=1$ in ①, we get

$$u_{i,2} = u_{i+1,1} + u_{i-1,1} - u_{i,0} \quad \text{for } i = 1,2,3,4$$

$$u_{1,2} = u_{2,1} + u_{0,1} - u_{1,0} = \frac{2}{3} + 0 - \frac{1}{3} = \frac{1}{3}$$

$$u_{2,2} = u_{3,1} + u_{1,1} - u_{2,0} = \frac{7}{12} + \frac{1}{3} - \frac{2}{3} = \frac{1}{4}$$

$$u_{3,2} = u_{4,1} + u_{2,1} - u_{3,0} = \frac{1}{2} + \frac{2}{3} - 1 = \frac{1}{6}$$

$$u_{4,2} = u_{5,1} + u_{3,1} - u_{4,0} = 0 + \frac{7}{12} - \frac{1}{2} = \frac{1}{12}$$

The above values constitute the second time level solution at $t = 2k = \frac{2}{10} = \frac{1}{5}$.

For the third time level we put $j=2$, in ①, we get

$$u_{i,3} = u_{i+1,2} + u_{i-1,2} - u_{i,1} \quad \text{for } i = 1,2,3,4$$

$$u_{1,3} = u_{2,2} + u_{0,2} - u_{1,1} = \frac{1}{4} + 0 - \frac{1}{3} = \frac{1}{12}$$

$$u_{2,3} = u_{3,2} + u_{1,2} - u_{2,1} = \frac{1}{6} + \frac{1}{3} - \frac{2}{3} = -\frac{1}{6}$$

$$u_{3,3} = u_{4,2} + u_{2,2} - u_{3,1} = \frac{1}{12} + \frac{1}{4} - \frac{7}{12} = -\frac{1}{4}$$

$$u_{4,3} = u_{5,2} + u_{3,2} - u_{4,1} = 0 + \frac{1}{6} - \frac{1}{2} = -\frac{1}{3}$$

Third time level solution

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For the Fourth time level we put $j=3$ in ①

$$u_{i,4} = u_{i+1,3} + u_{i-1,3} - u_{i,2} \quad \text{for } i=1,2,3,4$$

$$u_{1,4} = u_{2,3} + u_{0,3} - u_{1,2} = -\frac{1}{6} + 0 - \frac{1}{3} = -\frac{1}{2}$$

$$u_{2,4} = u_{3,3} + u_{1,3} - u_{2,2} = -\frac{1}{4} + \frac{1}{12} - \frac{1}{4} = -\frac{5}{12}$$

$$u_{3,4} = u_{4,3} + u_{2,3} - u_{3,2} = -\frac{1}{3} - \frac{1}{6} - \frac{1}{6} = -\frac{2}{3}$$

$$u_{4,4} = u_{5,3} + u_{3,3} - u_{4,2} = 0 - \frac{1}{4} - \frac{1}{12} = -\frac{1}{3}$$

The above values constitute the solution at fourth time level $t = 4k = 4\left(\frac{1}{10}\right) = \frac{2}{5}$.

⑤ The transverse displacement $u(x,t)$ of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $25\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial t^2} = 0$ with boundary conditions $u(0,t) = u(5,t) = 0$ and with initial conditions $u(x,0) = 20x$ when $0 \le x \le 1$ and $u(x,0) = 25\left(1 - \frac{x}{5}\right)$, $1 \le x \le 5$, $\frac{\partial u}{\partial t} = 0$ at $t=0$. Solve the equation numerically for $0 \le t \le 0.8$ take $h=1$.

Soln: Take $\beta = 1 \Rightarrow k = \frac{1}{5}$

t \ x (length)	0	1	2	3	4	5
0 (time level)	0	20	15	10	5	0
1 (1st time)	0	7.5	15	10	5	0
2 (2nd time)	0	−5	2.5	10	5	0
3 (3rd time)	0	−5	−10	−2.5	5	0
4	0	−5	−10	−15	−7.5	0

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